SMART CLASSROOM ENERGY MANAGEMENT SYSTEM

A MINI PROJECT-II REPORT

Submitted by

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in partial fulfillment for the award of the degree of

S.Y B.TECH

IN

COMPUTER SCIENCE ENGINEERING

KOLHAPUR INSTITUTE OF TECHNOLOGY'S

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(EMPOWERED AUTONOMOUS)

MAY 2025-26

KOLHAPUR INSTITUTE OF TECHNOLOGY'S
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CERTIFICATE

This is to certify that the Mini Project-II report entitled, **" SMART CLASSROOM ENERGY MANAGEMENT SYSTEM"** submitted by **"Aishwarya Patil, Sawita Pathania, Shweta Shinde, Sofiya Ladaji , Saniya Suryavanshi, Shivanjali Shinde"** (43, 61, 62, 63, 65, 70), in partial fulfillment for the award of the degree of **" S.Y B.TECH "** in **"COMPUTER SCIENCE AND ENGINEERING "** at Kolhapur Institute of Technology's College of Engineering Kolhapur (Empowered Autonomous), Kolhapur, Maharashtra, INDIA, is a record of his / her own work carried out under my / our supervision and guidance.

SIGNATURE	**SIGNATURE**
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HEAD OF THE DEPARTMENT	**SUPERVISOR**

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DECLARATION

We hereby declare that the Mini Project-II entitled, **"SMART CLASSROOM ENERGY MANAGEMENT SYSTEM"** submitted to Kolhapur Institute of Technology's College of Engineering Kolhapur (Empowered Autonomous), Kolhapur, Maharashtra, INDIA in the partial fulfillment of the award of the Degree of **"S.Y B.TECH"** in **"COMPUTER SCIENCE AND ENGINEERING"** is a Bonafide work carried out by me. The material contained in this Mini Project has not been submitted to any University or Institution for the award of any degree.

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ABSTRACT

Our project presents a **Smart Classroom Energy Management System**, developed to provide an intelligent and automated solution for reducing unnecessary electricity consumption in classrooms. The system uses the YOLO object detection model for real-time human detection and occupancy monitoring. A camera continuously captures classroom activity, and the system automatically detects the presence of people inside the classroom environment.

The project integrates Artificial Intelligence, Internet of Things (IoT), backend communication, and embedded systems to perform automatic device control. Based on occupancy detection, the backend server communicates with the ESP32 microcontroller through WiFi connectivity and controls classroom devices using relay modules. The system automatically switches devices such as lights and fans ON when a person is detected and turns them OFF when the classroom becomes empty, thereby improving energy efficiency and reducing electricity wastage.

In addition, the project includes a real-time frontend dashboard that displays occupancy status, device activity, and overall system operation in an interactive and user-friendly manner. The system demonstrates seamless integration between AI detection, IoT communication, frontend monitoring, and hardware automation technologies.

By combining modern technologies with smart automation concepts, the project aims to support sustainable energy management and smart classroom development. The Smart Classroom Energy Management System provides a low-cost, scalable, and efficient solution for intelligent classroom automation and future smart campus applications.

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CHAPTER 1

INTRODUCTION

The Mini project-II titled **"Smart Classroom Energy Management System"** is developed to provide an intelligent and automated solution for reducing unnecessary electricity consumption in classrooms. The system combines Artificial Intelligence and Internet of Things (IoT) technologies to automatically monitor classroom occupancy and control electrical devices such as lights and fans.

The project uses the YOLO (You Only Look Once) object detection model for real-time human detection inside the classroom. A camera continuously captures classroom video frames, and the YOLO model processes these frames to detect the presence of people. Based on occupancy detection, the backend server communicates with the ESP32 microcontroller through WiFi communication and controls devices using relay modules.

The Smart Classroom Energy Management System integrates AI detection, backend communication, embedded systems, and frontend monitoring into a single automation platform. The system helps reduce electricity wastage, improve energy efficiency, and support smart classroom automation using modern technologies.

1.1 PROJECT CATEGORY

The Smart Classroom Energy Management System belongs to the category of Artificial Intelligence and IoT-based automation systems. The project combines Computer Vision, Embedded Systems, Backend Communication, and Web Technologies to create an intelligent classroom environment.

The system uses the YOLO object detection model for occupancy detection and ESP32 microcontroller for automatic device control operations. The project also falls under the category of smart energy management systems because it focuses on reducing unnecessary electricity consumption through intelligent automation.

The project demonstrates the practical implementation of AI and IoT technologies for smart classroom and smart campus applications. It also supports sustainable energy management and modern infrastructure development.

1.2 OBJECTIVES

The main objective of the Smart Classroom Energy Management System is to reduce unnecessary electricity wastage through intelligent classroom automation. The project aims to automate classroom devices based on occupancy detection using Artificial Intelligence and IoT technologies.

The objectives of the project are as follows:

- To reduce unnecessary electricity consumption in classrooms.
- To automate classroom devices such as lights and fans.
- To detect human presence using the YOLO object detection model.
- To establish communication between backend server and ESP32 through WiFi.
- To control devices automatically using relay modules.

- To improve energy efficiency using intelligent automation.
- To provide real-time monitoring using a frontend dashboard.
- To demonstrate practical implementation of AI and IoT technologies.

The project also aims to provide a scalable and low-cost solution for smart classroom automation systems.

1.3 PROJECT FORMULATION

In traditional classroom environments, electrical devices such as lights and fans are manually controlled using switches. In many situations, these devices remain switched ON even when classrooms are empty. This results in unnecessary electricity wastage and increased power consumption.

The existing classroom system completely depends on human operation for controlling devices. Since there is no intelligent monitoring or occupancy detection system, there is a high possibility of negligence and inefficient energy management. Educational institutions often face increased electricity costs due to improper device management.

Traditional systems also lack automation, centralized monitoring, and real-time occupancy tracking capabilities. Therefore, there is a need for an intelligent classroom automation system that can automatically detect occupancy and control devices efficiently.
The Smart Classroom Energy Management System is developed to solve these problems using Artificial Intelligence and Internet of Things technologies.

1.4 IDENTIFICATION/RECOGNIZITION OF NEED

In modern educational institutions, effective energy management has become an important requirement. In many classrooms, electrical appliances such as lights, fans, and projectors remain switched ON even when classrooms are vacant. This leads to unnecessary electricity consumption, increased operational costs, and wastage of resources.

Traditional manual switching systems are not efficient because they completely depend on human operation. Sometimes users forget to switch OFF devices after leaving the classroom. Existing sensor-based systems also face problems such as inaccurate detection, limited coverage, and false triggering.

Therefore, there is a need for an intelligent and automated system that can accurately detect classroom occupancy and automatically control electrical appliances. The proposed Occupancy-Based Classroom Automation System fulfills this need by using AI-based human detection with the YOLO model and IoT-based device automation using ESP32 and relay modules. The system helps in reducing power wastage, improving automation efficiency, and supporting smart classroom infrastructure.

1.5 EXISTING SYSTEM

The existing classroom system is completely manual and depends on human operation for controlling electrical devices such as lights and fans. Users manually switch devices ON and OFF according to their requirements using traditional electrical switches.

In many classrooms, devices remain switched ON even when no students or faculty members are present inside the classroom. Since there is no occupancy detection mechanism, unnecessary electricity wastage occurs frequently. The traditional system lacks automation, intelligent decision-making, and centralized monitoring capabilities.

Existing systems do not provide real-time occupancy monitoring or automatic device control operations. Device management completely depends on user attention and responsibility, which reduces efficiency and increases operational costs.

The traditional classroom system is not suitable for modern smart infrastructure environments and requires intelligent automation technologies for efficient energy management.

1.6 Proposed System

The proposed Smart Classroom Energy Management System uses Artificial Intelligence and Internet of Things (IoT) technologies to automate classroom energy management based on occupancy detection.
The system continuously monitors classroom occupancy using a camera and the YOLO object detection model. The camera captures classroom video frames, and the YOLO model processes these frames in real time to detect human presence inside the classroom.

Once occupancy is detected, the information is transmitted to the backend server. The backend acts as the communication bridge between the AI detection module and the ESP32 microcontroller. Based on occupancy conditions, the backend sends ON/OFF commands to the ESP32 through WiFi communication.
The ESP32 microcontroller controls the relay module connected to classroom devices such as lights and fans. If a person is detected inside the classroom, the relay module automatically switches the connected devices ON. Similarly, when the classroom becomes empty, the devices are automatically switched OFF to reduce electricity wastage.

The project also includes a frontend dashboard for real-time monitoring of occupancy status, device activity, and system operation. The proposed system improves energy efficiency, reduces manual effort, and supports intelligent classroom automation using modern technologies.

1.7 Unique Features

The Occupancy-Based Classroom Automation System provides several advanced and unique features compared to traditional automation systems.

- The system uses the YOLO object detection model for accurate real-time human detection.
- It automatically controls classroom appliances based on occupancy status without requiring human intervention.
- The system integrates IoT technology using ESP32 and relay modules for efficient communication and device control.
- A real-time dashboard is provided to display occupancy status and appliance control information.
- The system reduces unnecessary electricity consumption and improves energy efficiency.
- Continuous monitoring of classroom occupancy is performed using camera-based computer vision technology.
- The proposed system provides faster response time and more accurate detection compared to traditional sensor-based systems.

CHAPTER 2

REQUIREMENT ANALYSIS AND SYSTEM SPECIFICATION

The Requirement Analysis and System Specification phase plays an important role in the development of the Smart Classroom Energy Management System. This phase focuses on identifying the hardware requirements, software requirements, operational feasibility, expected challenges, and development methodology required for successful implementation of the project.

The Smart Classroom Energy Management System combines Artificial Intelligence, Internet of Things (IoT), backend communication, and embedded systems to perform intelligent classroom automation. Proper requirement analysis ensures smooth integration between software and hardware modules and helps achieve efficient system performance.

The system requirements include AI-based occupancy detection, backend communication, WiFi-enabled microcontroller integration, relay switching operations, and frontend monitoring functionality. The project is designed to provide a low-cost, scalable, and energy-efficient solution for smart classroom automation.

2.1 Feasibility Study

The feasibility study is conducted to determine whether the Smart Classroom Energy Management System is technically, economically, and operationally feasible for implementation.

Technical Feasibility

The project is technically feasible because all required technologies and hardware components are easily available and compatible with each other. Technologies such as YOLO object detection, ESP32 microcontroller, relay modules, backend communication frameworks, and frontend web technologies can be integrated successfully to create an intelligent automation platform.
The YOLO object detection model provides accurate real-time occupancy detection, while ESP32 enables efficient IoT communication and device control operations. The frontend dashboard and backend server ensure smooth communication and monitoring functionality.

Economic Feasibility

The project is economically feasible because the hardware components used in the system are low-cost and easily available in the market. Components such as ESP32, relay modules, LEDs, breadboards, jumper wires, and cameras are affordable and suitable for prototype implementation.
The software tools used in the project, such as Python, Flask, Arduino IDE, HTML, CSS, and JavaScript, are open-source or freely available. Therefore, the overall implementation cost of the project is low and suitable for educational institutions.

Operational Feasibility

The project is operationally feasible because the system is simple, user-friendly, and easy to operate. The frontend dashboard provides real-time monitoring of occupancy status and device activity, reducing manual effort and improving classroom energy management efficiency.
The automation process reduces human dependency for controlling classroom devices and improves overall operational efficiency. The system also supports future scalability and smart infrastructure integration.

2.2 Software Requirement Specification

The Smart Classroom Energy Management System requires both software and hardware components for successful implementation and operation.

Software Requirements

The software tools used in the project are:

- Python
- YOLO Object Detection Model
- Flask / Node.js
- Arduino IDE
- HTML
- CSS
- JavaScript
- Visual Studio Code (VS Code)

Python is used for implementing the YOLO object detection model and processing classroom video frames. YOLO performs real-time occupancy detection using Computer Vision techniques.
Flask or Node.js is used for backend communication between the AI detection module and ESP32 microcontroller. The backend server processes occupancy information and sends ON/OFF commands through WiFi communication.
HTML, CSS, and JavaScript are used for frontend dashboard development and real-time monitoring functionality. Arduino IDE is used for programming the ESP32 microcontroller and implementing relay control operations.

Hardware Requirements

The hardware components used in the project are:

- ESP32 Microcontroller
- Relay Module
- LED
- Breadboard
- Jumper Wires

- USB Cable
- Camera
- Power Supply

The ESP32 acts as the IoT controller responsible for receiving backend commands and controlling relay operations. The relay module performs automatic switching of classroom devices.
In the current prototype implementation, an LED is used to represent classroom devices such as lights and fans. The camera continuously captures classroom video frames for occupancy detection.

2.3 Validation

Validation is performed to ensure that all modules and functionalities of the Smart Classroom Energy Management System operate correctly and efficiently.

The YOLO object detection model is tested under different occupancy conditions to verify accurate human detection performance. The backend communication module is validated to ensure successful transmission of occupancy information and ON/OFF commands between the backend server and ESP32 microcontroller.

The ESP32 microcontroller is tested for proper WiFi connectivity and successful response to backend commands. Relay modules and LED prototype operations are validated to ensure correct device switching functionality.

The frontend dashboard is also tested for real-time monitoring and successful display of occupancy status and device activity. Complete integration testing confirms successful communication between all software and hardware modules.

The validation process ensures stable system performance, reliable automation functionality, and efficient occupancy-based device control operations.

2.4 Expected Hurdles

Several challenges and difficulties may occur during the development and implementation of the Smart Classroom Energy Management System.

One of the major expected hurdles is WiFi connectivity issues between the backend server and ESP32 microcontroller. Communication interruptions may affect real-time device control operations and automation efficiency.

The YOLO object detection model may face detection accuracy limitations under low lighting conditions, unclear camera visibility, or multiple occupancy situations. Proper camera positioning and lighting conditions are required for accurate occupancy detection.

Hardware synchronization problems, relay operation delays, ESP32 upload issues, and backend communication errors may also occur during implementation. Integration between software and hardware modules requires continuous testing and debugging.

Another expected challenge is maintaining smooth real-time communication between AI processing, backend server, and IoT hardware modules. Proper optimization and error handling techniques are necessary to ensure stable system operation.

Despite these challenges, continuous testing, debugging, and optimization help achieve reliable automation functionality and successful project implementation.

2.5 SDLC Model

The Smart Classroom Energy Management System follows the Software Development Life Cycle (SDLC) model for systematic project development and implementation.

Requirement Analysis Phase

This phase focuses on identifying project objectives, system requirements, hardware components, software tools, and automation functionalities required for project implementation.

System Design Phase

The system architecture, communication workflow, frontend design, backend structure, and hardware integration process are designed during this phase. Flowcharts, architecture diagrams, and circuit diagrams are also prepared.

Implementation Phase

The implementation phase focuses on developing the YOLO object detection model, backend server, frontend dashboard, ESP32 programming, and relay module integration.

Testing Phase

Testing is performed on all software and hardware modules to ensure proper communication and successful automation functionality. Integration testing validates complete system operation.

Deployment and Maintenance Phase

The final system is demonstrated after successful implementation and testing. Future improvements, debugging, scalability enhancements, and maintenance activities are also considered during this phase.
The SDLC model helps ensure systematic development, proper documentation, organized workflow, and successful integration of all project modules.

CHAPTER 3

SYSTEM DESIGN

The Smart Classroom Energy Management System is designed to perform intelligent classroom automation using Artificial Intelligence and Internet of Things (IoT) technologies. The system integrates multiple software and hardware modules that work together efficiently to perform occupancy-based device control operations.

The system mainly consists of the YOLO Detection Module, Backend Communication Module, ESP32 Control Module, Relay Switching Module, and Frontend Dashboard Module. These modules communicate with each other through WiFi and backend processing to create a complete automation platform.

The system is designed to continuously monitor classroom occupancy using a camera and automatically control classroom devices such as lights and fans. The modular architecture improves scalability, maintainability, and future enhancement capabilities.

3.1 Design Approach

The Smart Classroom Energy Management System follows a modular and function-oriented design approach. Each module of the system performs a specific operation while maintaining communication with other modules.

The YOLO Detection Module is responsible for detecting human presence inside the classroom using real-time video processing. The Backend Communication Module receives occupancy information from the YOLO model and sends control commands to the ESP32 microcontroller through WiFi communication.

The ESP32 Control Module processes the received commands and controls the relay module connected to classroom devices. The Frontend Dashboard Module displays occupancy status, device activity, and system information in real time.

This modular design improves system flexibility and simplifies future system expansion and maintenance operations.

3.2 Detailed Design

The operation of the Smart Classroom Energy Management System begins with video capture through a classroom camera. The camera continuously captures classroom video frames and sends them to the YOLO object detection model for processing.

The YOLO model analyzes video frames and detects whether a person is present inside the classroom environment. Once occupancy is detected, the backend server receives occupancy information from the AI detection module.

The backend server acts as the communication bridge between the YOLO detection system and the ESP32 microcontroller. Based on occupancy conditions, the backend generates ON/OFF commands and transmits them to the ESP32 through WiFi communication.

The ESP32 microcontroller receives the commands and controls the relay module connected to classroom devices. In the prototype implementation, an LED is used to represent classroom devices such as lights and fans.

The frontend dashboard continuously displays occupancy status, device activity, and backend communication updates in real time.

3.3 Structured Design Tools

The Smart Classroom Energy Management System uses various structured design tools for representing system workflow, communication process, and hardware integration.

Flowchart

The flowchart represents the complete automation process of the system. It shows the sequence of occupancy detection, backend communication, ESP32 processing, relay operation, and automatic device control.

Architecture Diagram

The architecture diagram represents communication between the camera, YOLO model, backend server, ESP32 microcontroller, relay module, and frontend dashboard.

Circuit Diagram

The circuit diagram shows hardware connections between ESP32, relay module, LED, breadboard, and power supply.

Data Flow Diagram (DFD)

The DFD represents the flow of occupancy information from the camera to the backend server and communication between different modules.
These structured design tools help improve understanding of system functionality and communication workflow.

3.4 User Interface Design

The frontend dashboard of the Smart Classroom Energy Management System is developed using HTML, CSS, and JavaScript. The user interface provides real-time monitoring of occupancy detection, device activity, and overall system operation.

The dashboard displays whether a person is detected inside the classroom and whether classroom devices are currently ON or OFF. Proper indicators and status messages are used to improve user interaction and visualization.

The user interface is designed to be simple, responsive, and easy to use. The dashboard helps users monitor automation operations efficiently and improves accessibility of system information.

The frontend also supports future scalability and additional monitoring features such as energy consumption tracking and cloud-based device management.

3.5 Database Design

The Smart Classroom Energy Management System uses backend storage for managing occupancy records, device activity logs, timestamps, and communication data.

The database structure stores information related to:

- Occupancy detection status
- Device ON/OFF activity
- Backend communication logs
- Timestamp records
- System monitoring information

The database design supports future enhancements such as:

- Energy monitoring
- Historical occupancy analysis
- Smart analytics
- Cloud connectivity
- Multiple classroom management

The backend database structure is designed to minimize redundancy and improve efficient data retrieval operations.

3.6 Methodology

The methodology of the Smart Classroom Energy Management System combines Artificial Intelligence, backend communication, IoT hardware, and frontend technologies to create intelligent automation.
The system workflow follows this sequence:

Camera → YOLO Detection → Backend Server → ESP32 → Relay Module → Device Control
The camera continuously captures classroom video frames. The YOLO object detection model processes these frames and detects human presence inside the classroom.

The backend server receives occupancy information and sends ON/OFF commands to the ESP32 through WiFi communication. The ESP32 controls the relay module connected to classroom devices such as lights and fans.

The frontend dashboard continuously updates occupancy status and device activity for real-time monitoring. This methodology ensures smooth integration between AI processing, backend communication, and IoT automation technologies.

3.7 System Design Flowchart

The system flowchart represents the complete workflow of occupancy-based classroom automation.
The process begins with system initialization and video capture through the classroom camera. The YOLO model processes video frames and checks for human presence.

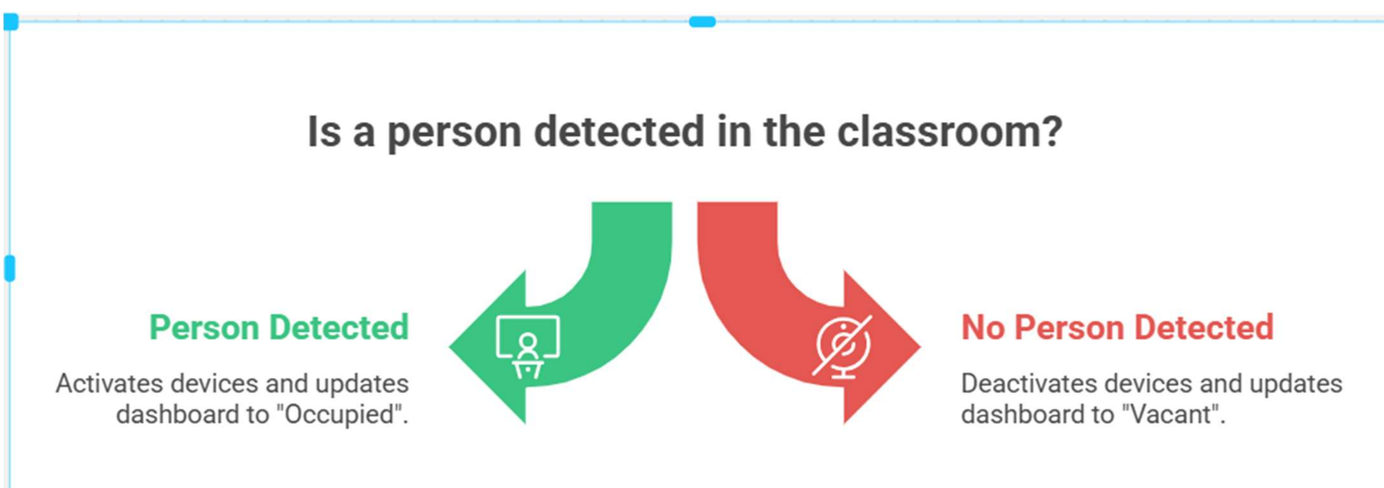
If a person is detected:

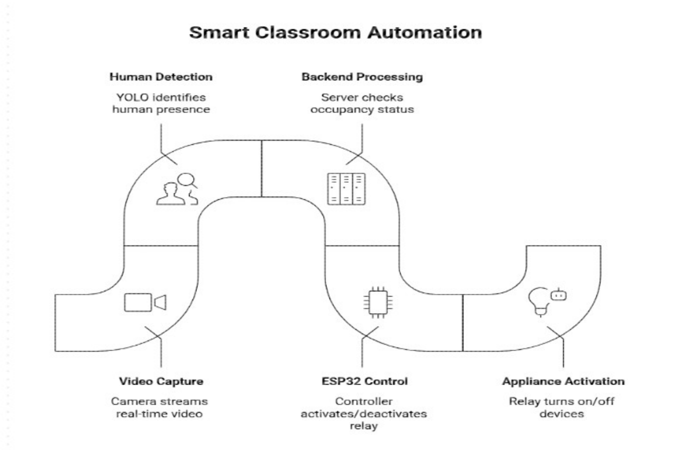
- Backend receives occupancy information
- ESP32 receives ON command
- Relay module switches devices ON
- Frontend dashboard updates occupancy status

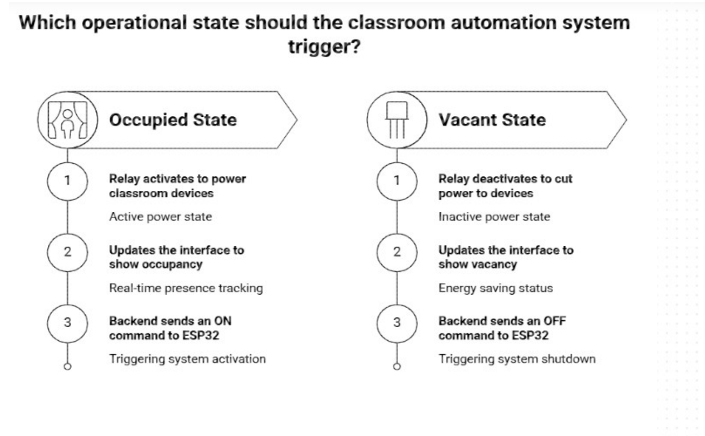
If no person is detected:

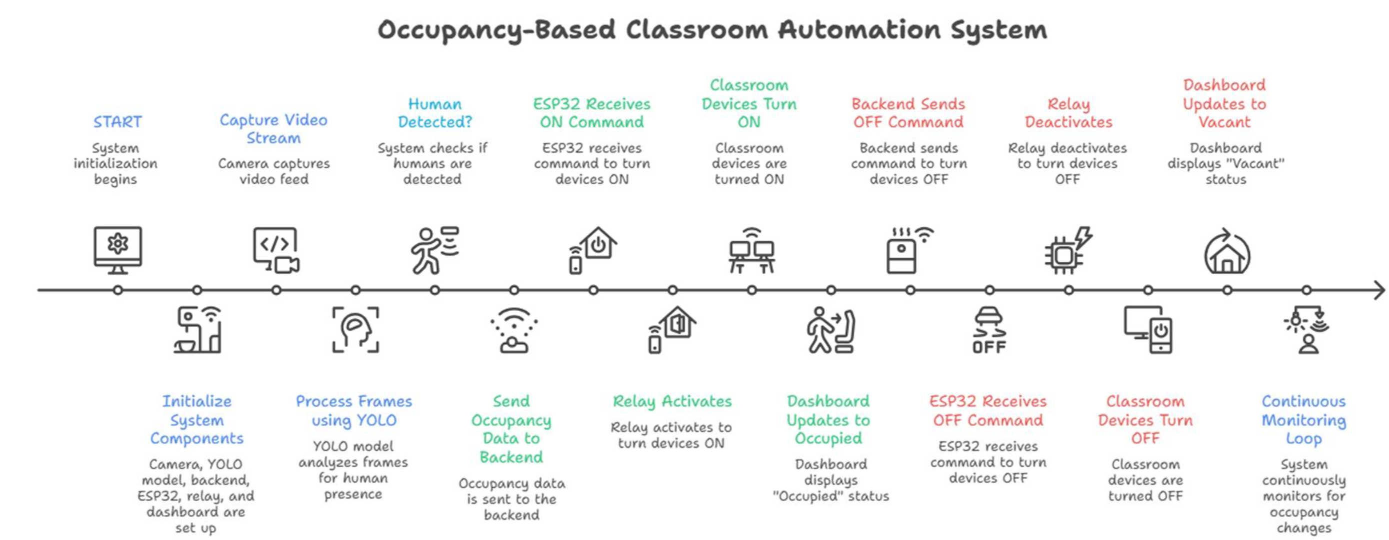
- Backend sends OFF command
- ESP32 switches relay OFF
- Devices turn OFF automatically
- Dashboard updates system status

The flowchart helps visualize system operation clearly and demonstrates communication between all system modules.




Smart Classroom Automation
Human Detection
YOLO identifies human presence
Backend Processing
Server checks occupancy status
Video Capture
Camera streams real-time video
ESP32 Control
Controller activates/deactivates relay
Appliance Activation
Relay turns on/off devices


Which operational state should the classroom automation system trigger?
Occupied State
1
Relay activates to power classroom devices
Active power state
2
Updates the interface to show occupancy
Real-time presence tracking
3
Backend sends an ON command to ESP32
Triggering system activation
Vacant State
1
Relay deactivates to cut power to devices
Inactive power state
2
Updates the interface to show vacancy
Energy saving status
3
Backend sends an OFF command to ESP32
Triggering system shutdown


Occupancy-Based Classroom Automation System
START
System initialization begins
Initialize System Components
Camera, YOLO model, backend, ESP32, relay, and dashboard are set up
Capture Video Stream
Camera captures video feed
Process Frames using YOLO
YOLO model analyzes frames for human presence
Human Detected?
System checks if humans are detected
Send Occupancy Data to Backend
Occupancy data is sent to the backend
ESP32 Receives ON Command
ESP32 receives command to turn devices ON
Relay Activates
Relay activates to turn devices ON
Classroom Devices Turn ON
Classroom devices are turned ON
Dashboard Updates to Occupied
Dashboard displays "Occupied" status
Backend Sends OFF Command
Backend sends command to turn devices OFF
ESP32 Receives OFF Command
ESP32 receives command to turn devices OFF
Relay Deactivates
Relay deactivates to turn devices OFF
Classroom Devices Turn OFF
Classroom devices are turned OFF
Dashboard Updates to Vacant
Dashboard displays "Vacant" status
Continuous Monitoring Loop
System continuously monitors for occupancy changes
OFF

CHAPTER 4

IMPLEMENTING,TESTING AND MAINTENANCE

The implementation phase of the Smart Classroom Energy Management System focuses on integrating Artificial Intelligence, backend communication, embedded systems, and frontend monitoring technologies into a complete automation platform.

The system is implemented using software tools such as Python, YOLO, Flask/Node.js, HTML, CSS, JavaScript, and Arduino IDE. Hardware components such as ESP32, relay modules, LEDs, and breadboards are integrated successfully for intelligent device control operations.

Proper testing and debugging techniques are performed to ensure smooth communication and successful automation functionality.

4.1 Implementation Tools

The Smart Classroom Energy Management System is implemented using multiple software and hardware tools.

Software Tools

- Python
- YOLO Object Detection Model
- Flask / Node.js
- Arduino IDE
- HTML
- CSS
- JavaScript
- VS Code

Python is used for implementing the YOLO object detection model and processing classroom video frames. Flask or Node.js is used for backend communication between the AI detection module and ESP32 microcontroller.
HTML, CSS, and JavaScript are used for frontend dashboard development and real-time monitoring functionality.

Hardware Tools

- ESP32 Microcontroller
- Relay Module
- LED
- Breadboard
- Jumper Wires
- USB Cable
- Camera

The ESP32 acts as the IoT controller responsible for receiving backend commands and controlling relay operations.

4.2 Coding Standards

Proper coding standards are followed during project implementation to improve software quality, readability, and maintainability.

Meaningful variable names and modular programming concepts are used throughout the project. Proper indentation, formatting, and commenting techniques are followed to improve understanding of source code. Error handling techniques are implemented for backend communication and hardware synchronization.

Separate modules are created for occupancy detection, backend communication, ESP32 programming, and frontend monitoring to simplify debugging and future enhancement operations.

These coding standards improve software reliability and simplify system maintenance.

4.3 Project Scheduling

The development of the Smart Classroom Energy Management System is divided into multiple phases to ensure systematic implementation and successful completion.

Project Development Phases

1. Requirement Analysis
2. Hardware Component Selection
3. YOLO Model Implementation
4. Backend Communication Development
5. ESP32 Programming
6. Frontend Dashboard Design
7. Hardware Integration
8. System Testing and Debugging

9. Final System Demonstration

Proper project scheduling improves coordination between software and hardware development tasks and helps achieve project objectives within the planned timeline.

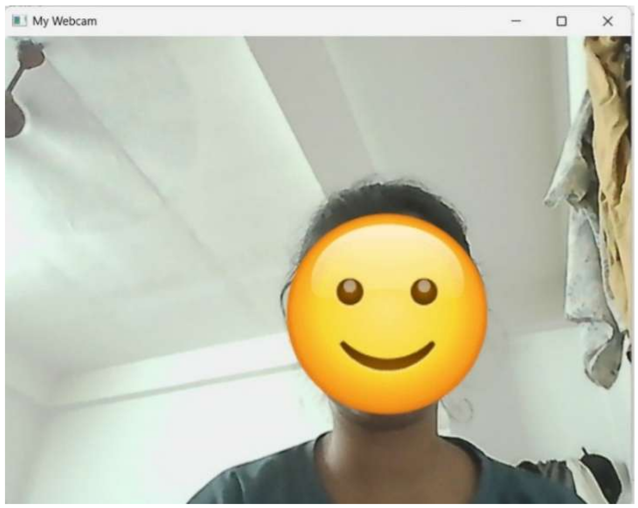
4.4 Testing

Testing is performed on all modules of the Smart Classroom Energy Management System to ensure successful automation functionality and communication reliability.

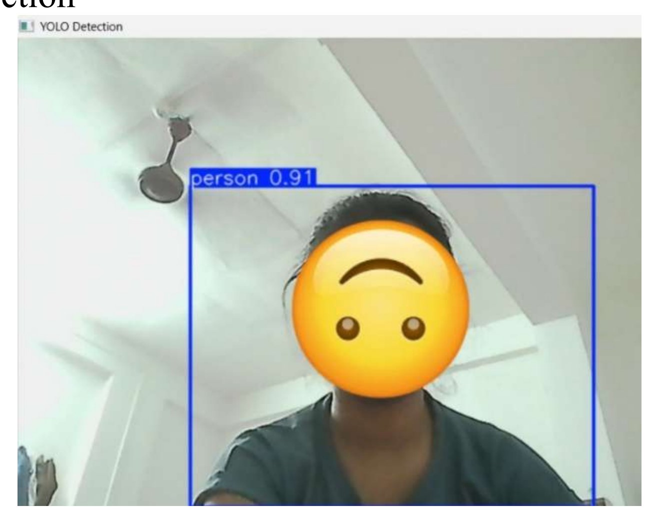
YOLO Detection Testing

The YOLO object detection model is tested for accurate human detection under different classroom conditions.

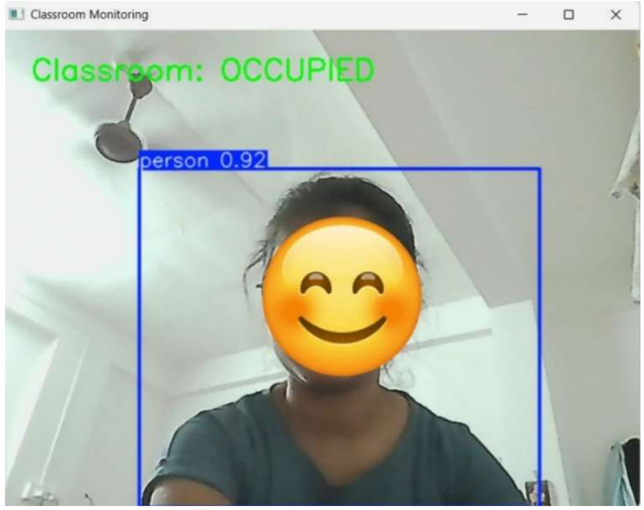
1. Webcam access



2. YOLO detection

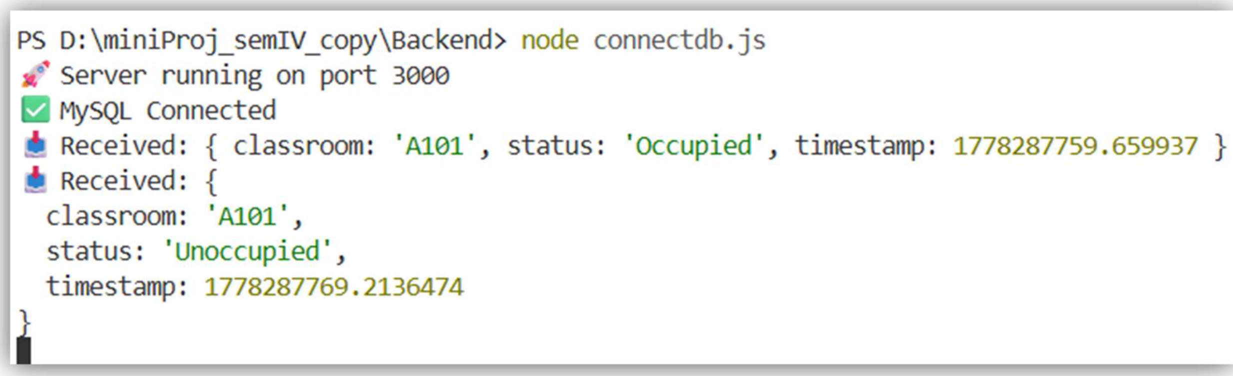


3. Classroom status



Backend Communication Testing

Backend communication is tested to ensure successful transmission of occupancy information and ON/OFF commands to the ESP32 microcontroller.

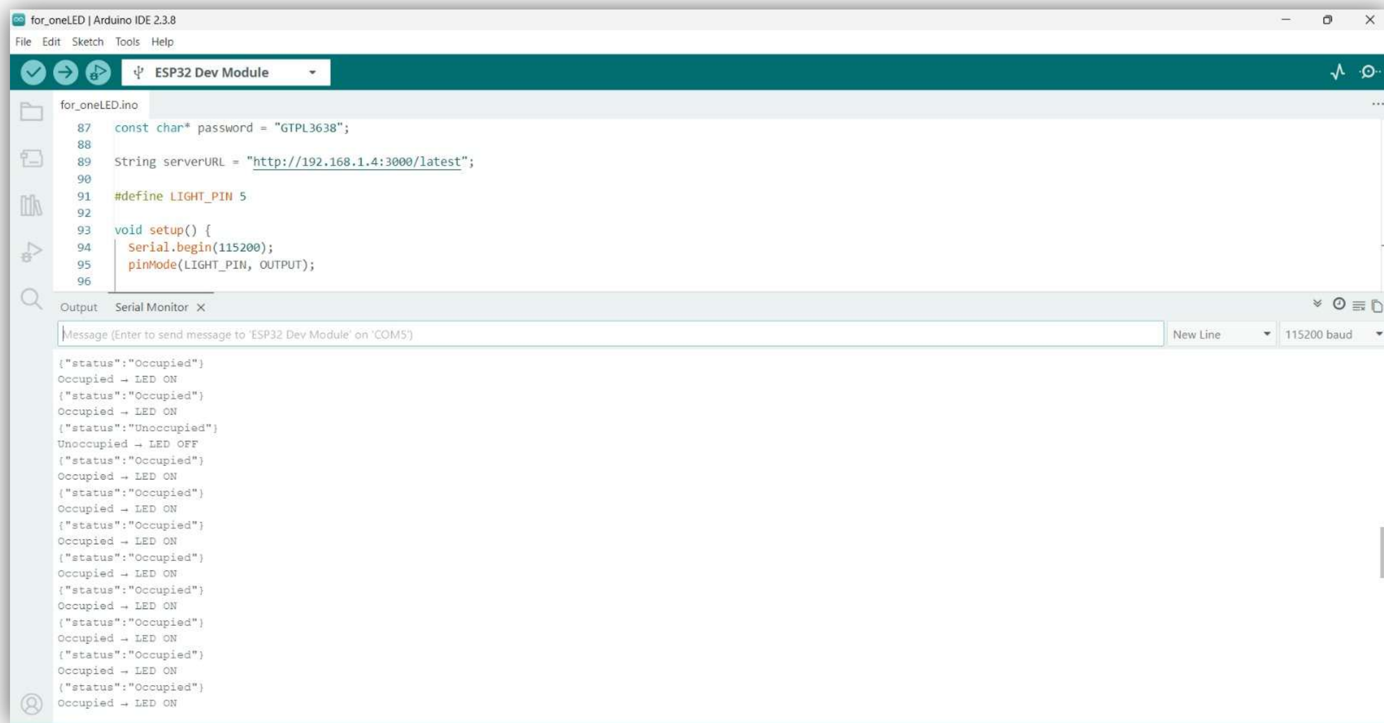


```
PS D:\miniProj_semIV_copy\Backend> node connectdb.js
🚀 Server running on port 3000
✅ MySQL Connected
📥 Received: { classroom: 'A101', status: 'Occupied', timestamp: 1778287759.659937 }
📥 Received: {
  classroom: 'A101',
  status: 'Unoccupied',
  timestamp: 1778287769.2136474
}
```

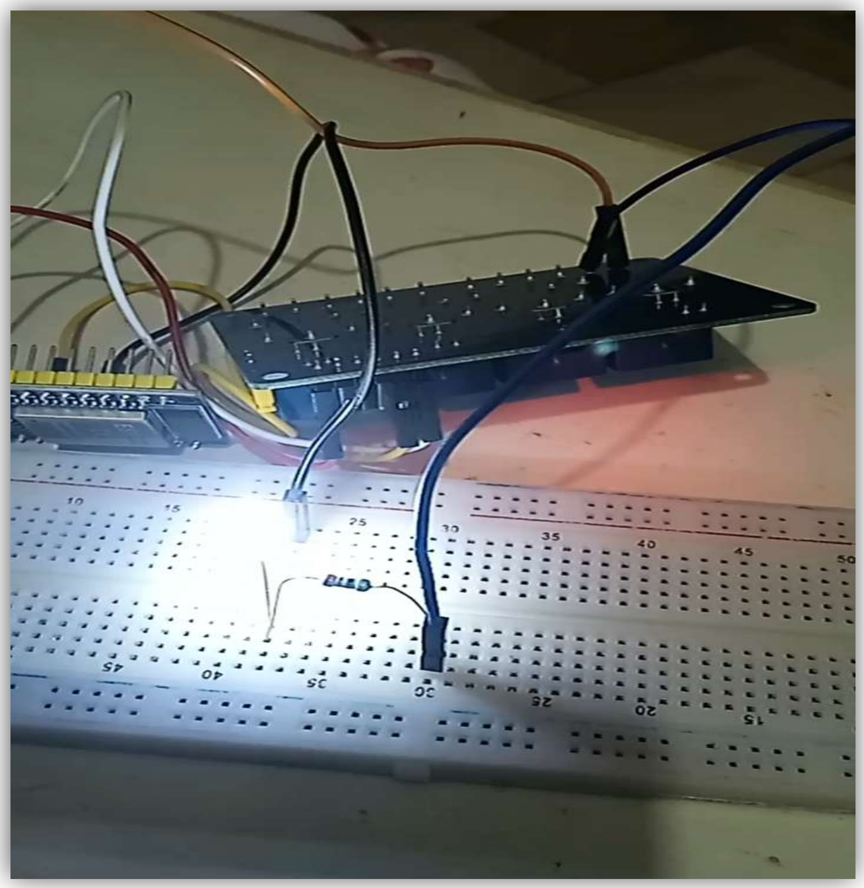
ESP32 and Relay Testing

The ESP32 microcontroller and relay module are tested for proper device switching operations.

1. ESP32 Arduino IDE



2. Hardware Part Testing

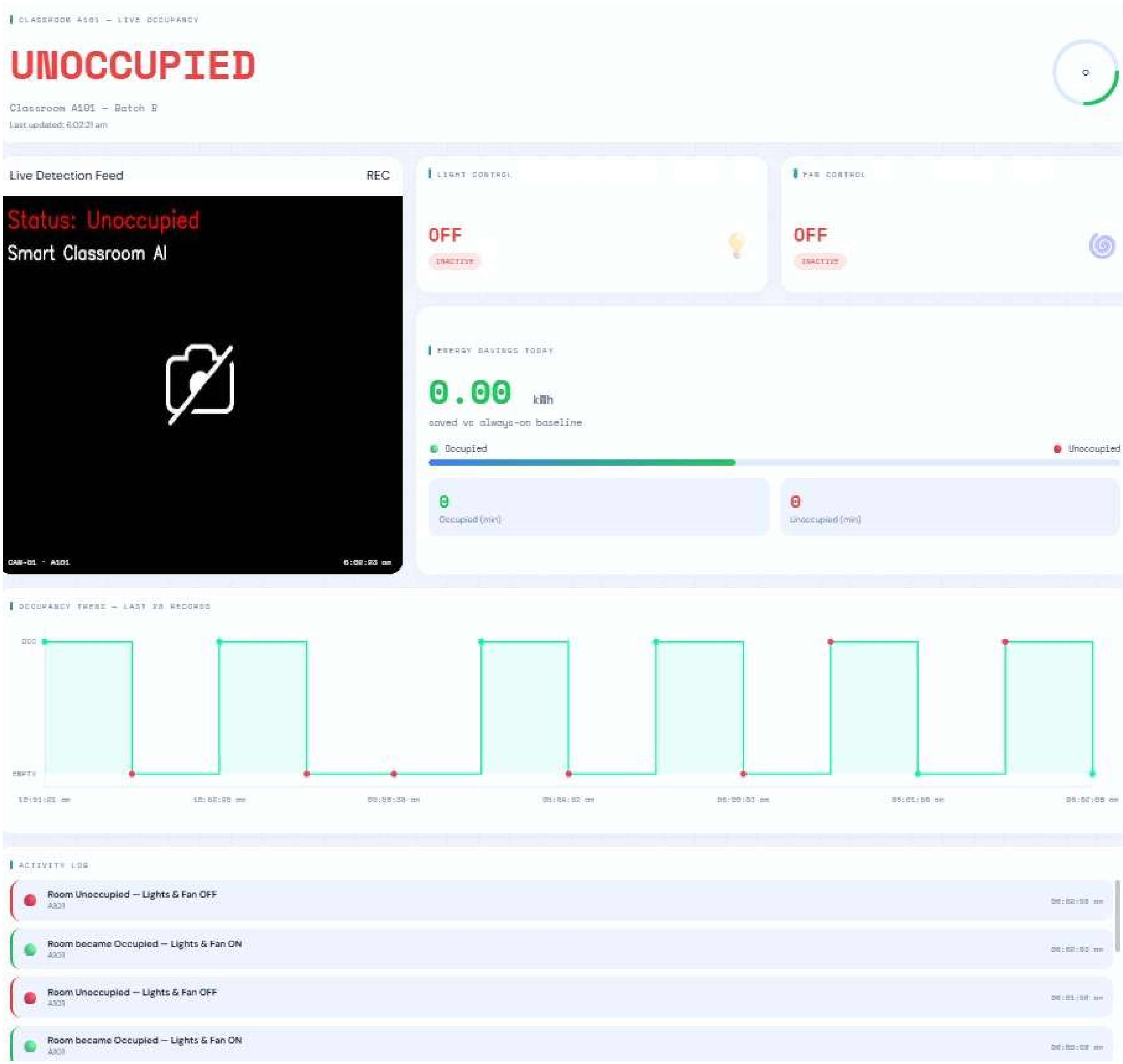


Frontend Dashboard Testing

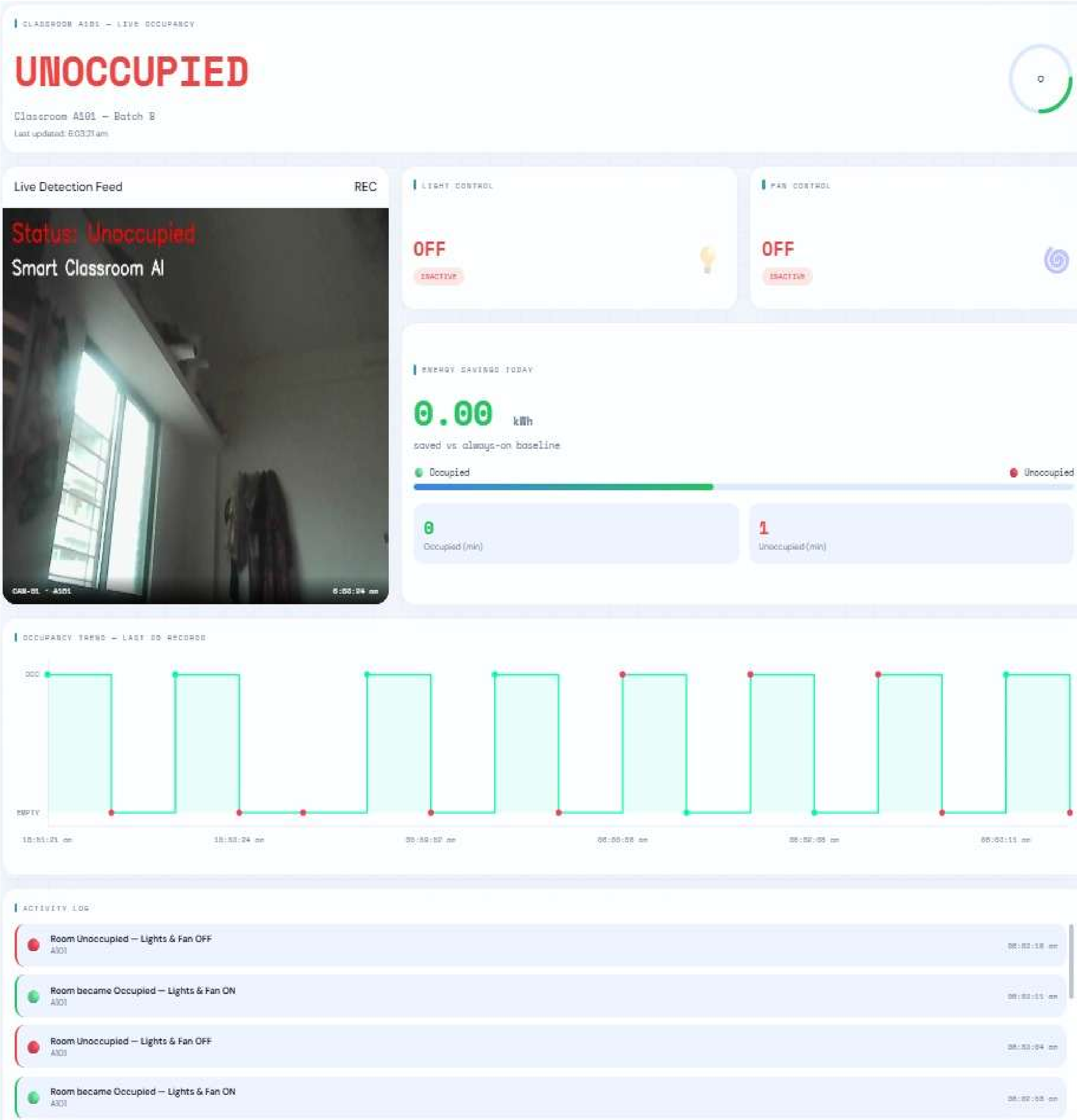
The frontend dashboard is tested for real-time monitoring and successful display of occupancy and device status information.
Complete integration testing confirms successful communication between all modules and stable automation performance.

1. Status : Occupied

2. Webcam closed



3. Status: Empty



CHAPTER 5

RESULTS AND DISCUSSION

The Smart Classroom Energy Management System successfully demonstrates occupancy-based intelligent classroom automation using Artificial Intelligence and Internet of Things technologies.

The project integrates YOLO object detection, backend communication, ESP32 automation, relay switching, and frontend monitoring into a complete working prototype. The system successfully performs automatic device control based on classroom occupancy conditions.

The implementation results confirm successful communication between software and hardware modules and demonstrate practical application of AI and IoT technologies for smart classroom automation.

5.1 User Interface Representation

The user interface of the Occupancy-Based Classroom Automation System is designed to provide simple and real-time monitoring of classroom occupancy and device status. The dashboard helps the user easily understand whether the classroom is occupied or vacant.

The interface displays the live occupancy status detected by the YOLO model. If a person is detected in the classroom, the dashboard shows the status as **Occupied** and the connected devices such as lights and fans are shown as **ON**. If no person is detected, the dashboard updates the status as **Vacant** and the devices are shown as **OFF**.

The dashboard also displays important system information such as camera status, ESP32 connection status, relay status, and real-time detection updates. This helps the user verify that all system modules are working properly.

The user interface is designed to be clear, simple, and user-friendly. It reduces the need for manual checking and allows the system administrator or teacher to monitor classroom automation easily.

5.2 Module Description

The Smart Classroom Energy Management System consists of multiple modules that work together to perform intelligent classroom automation and occupancy-based device control.

YOLO Detection Module

The YOLO Detection Module is responsible for detecting human presence inside the classroom using real-time video processing. The module continuously analyzes video frames captured through a camera and identifies whether a person is present inside the classroom environment.

Backend Communication Module

The Backend Communication Module acts as the communication bridge between the AI detection system and the ESP32 microcontroller. It receives occupancy information from the YOLO model, processes the data, and sends appropriate ON/OFF commands to the ESP32 through WiFi communication.

ESP32 Control Module

The ESP32 Control Module receives backend commands and controls the relay module connected to classroom devices. The ESP32 performs automatic switching operations based on occupancy conditions.

Relay Switching Module

The Relay Switching Module controls electrical devices such as lights and fans according to the commands received from the ESP32 microcontroller. In the prototype implementation, an LED is used to represent classroom devices.

Frontend Dashboard Module

The Frontend Dashboard Module provides real-time monitoring of occupancy detection, device activity, and overall system status. The dashboard improves user interaction and visualization of automation operations. All modules work together efficiently to create a complete smart classroom automation system.

5.3 Snapshots of System

The Smart Classroom Energy Management System includes various software and hardware snapshots that demonstrate successful implementation and operation of the project.

The frontend dashboard snapshots display occupancy detection status and device activity in real time. These screenshots demonstrate successful communication between the frontend interface and backend server. The backend server snapshots demonstrate successful processing of occupancy information and communication with the ESP32 microcontroller.

The hardware setup snapshots include ESP32 connections, relay module integration, breadboard setup, jumper wire connections, and LED prototype implementation. These images demonstrate successful integration of IoT hardware components.

The system snapshots collectively confirm successful implementation and working of the Smart Classroom Energy Management System.

5.4 Back-End Representation

The backend server plays an important role in the Smart Classroom Energy Management System because it acts as the communication bridge between the YOLO object detection model and ESP32 microcontroller. The backend continuously receives occupancy data from the YOLO model. Once a person is detected, the backend processes the information and generates appropriate ON/OFF commands for the ESP32 microcontroller.

These commands are transmitted through WiFi communication for automatic device control operations. The backend also handles system synchronization and frontend status updates.

The backend architecture supports future scalability and additional features such as:

- Energy monitoring
- Cloud connectivity
- Multiple classroom management
- Data logging
- Smart analytics

The successful backend implementation confirms reliable communication and efficient integration between Artificial Intelligence, IoT hardware, and frontend monitoring systems.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The Smart Classroom Energy Management System successfully achieves intelligent classroom automation and energy-efficient device management using Artificial Intelligence and Internet of Things technologies.

The project demonstrates automatic occupancy detection using the YOLO object detection model and automatic device control using ESP32 and relay modules. The integration of frontend dashboard, backend communication, AI detection model, and IoT hardware creates a complete automation platform.

The system automatically switches classroom devices ON when occupancy is detected and turns them OFF when the classroom becomes empty. This intelligent automation process helps reduce unnecessary electricity wastage and improves energy efficiency.

The project also demonstrates practical implementation of Computer Vision, IoT communication, embedded systems, backend integration, and frontend technologies for smart classroom automation.

Overall, the Smart Classroom Energy Management System provides a low-cost, scalable, and efficient solution for intelligent classroom automation and sustainable energy management.

6.2 Future Scope

The Smart Classroom Energy Management System can be further improved by integrating additional technologies and advanced **6**automation features.

Future versions of the project can control multiple classroom appliances such as:

- Lights
- Fans
- Projectors
- Air Conditioners
- Smart Boards

Real-time energy monitoring systems can also be integrated to measure electricity consumption and display energy-saving statistics through the frontend dashboard.

Cloud connectivity can enable remote monitoring and centralized control of multiple classrooms through internet-based services. Mobile applications can also be developed for remote device management and monitoring.

Additional sensors such as:

- Temperature Sensors
- PIR Sensors
- Light Intensity Sensors
- Environmental Monitoring Sensors

can improve automation accuracy and intelligent decision-making.

Future improvements may also include:

- Smart Analytics
- AI-based Predictive Energy Management
- Voice Control Integration
- Automated Scheduling Systems
- Smart Campus Integration

The project architecture is scalable and flexible, making it suitable for future smart automation and smart infrastructure applications.

REFERENCES / BIBLIOGRAPHY

In order to successfully complete the Smart Classroom Energy Management System, various technical resources, official documentations, research papers, and online references were studied throughout the development process. These references provided valuable guidance for implementing technologies such as YOLO object detection, ESP32 microcontroller programming, backend communication, IoT automation, and frontend dashboard development.

Official documentation and tutorials helped in understanding real-time occupancy detection, WiFi communication, relay module integration, and embedded system programming. Research papers and IoT-based automation articles provided insights into smart energy management systems and intelligent automation techniques.

Additional resources such as developer communities, online forums, and technical blogs were also referred to for troubleshooting hardware integration issues, backend communication errors, and optimization of AI detection models. The combination of official documentation and practical development resources ensured both theoretical understanding and successful implementation of the project.

Furthermore, various educational articles and IEEE research papers related to Artificial Intelligence, IoT systems, and smart automation technologies were used to improve the project architecture and functionality. These references helped maintain technical accuracy, system reliability, and proper implementation standards throughout the project development process.

List of References

1.YOLO Official Documentation – https://docs.ultralytics.com
Provided detailed guidance for implementing the YOLO object detection model, real-time human detection, and Computer Vision processing used in the Smart Classroom Energy Management System.

2.ESP32 Official Documentation – https://www.espressif.com

Provided information regarding ESP32 microcontroller programming, WiFi communication, GPIO operations, and IoT-based device control implementation.

3.Arduino IDE Documentation – https://www.arduino.cc

Used for understanding ESP32 programming, embedded system development, serial communication, and relay module integration using Arduino IDE.

4.Python Official Documentation – https://www.python.org

Provided reference materials for Python programming, backend processing, AI model integration, and communication handling within the project.

5.Flask Documentation – https://flask.palletsprojects.com

Helped in developing the backend communication server and handling occupancy data transfer between the YOLO model and ESP32 microcontroller.

6.OpenCV Documentation – https://opencv.org

Provided guidance for image processing, video frame handling, and real-time camera operations used in occupancy detection.

7.HTML, CSS, and JavaScript Documentation – https://developer.mozilla.org

Used for frontend dashboard development, responsive user interface design, and real-time monitoring implementation.

8.Node.js Documentation – https://nodejs.org

Provided backend communication concepts and server-side processing techniques for real-time system communication.

10.IEEE Research Papers on Smart Energy Management Systems
Provided technical insights regarding intelligent automation, energy-efficient systems, and smart classroom applications using AI and IoT technologies.

10.Research Articles on IoT-Based Automation Systems

Used for understanding IoT communication architecture, smart device control techniques, and automation methodologies.

10.Embedded Systems and IoT Reference Books

Provided theoretical knowledge related to microcontrollers, embedded programming, sensor integration, and relay-based automation systems.

12.Artificial Intelligence and Computer Vision Research Papers

Used for studying object detection algorithms, AI-based occupancy monitoring, and real-time detection methodologies.

13.Smart Classroom Automation Research Articles

Provided implementation ideas and architecture references for intelligent classroom automation and occupancy-based energy management systems.

14.Web Development and Backend Integration Documentation

Used for frontend-backend integration, API communication, real-time data synchronization, and dashboard implementation.